

STA 5320 - Project 1
Team 1
Daysha Watts
Joshua Martinez
Sharadha Varadharaj

Part 1

Convert both series into a tsibble. Make a plot of both series and comment on what you see.

Comment on the plots:

Original BJsales Data: Based on the plot it clearly indicates that there is an upward trend overall. The variability in the series does exist in the series. The series turns out to be non-stationary. It is also observed that there is a drastic increase in sales between 75 to 100 series. To comment on seasonality, there are no seasonal components in the series.

Original BJsales.lead Data: The plot definitely shows an upward trend similar to the BJsales data. The data seems to be non-stationary due to the presence of trend. There is variability in the series. It does not show a seasonal component in the data.

```
library(fpp3)
library(tsibble)
library(ggplot2)
library(forecast)
library(feasts)
library(patchwork)

# Loading the datasets
data("BJsales")
data("BJsales.lead")

# Subsetting the data of 130 observations
train_bjsales <- BJsales[1:130]
test_bjsales <- BJsales[131:150]

train_bjsales_lead <- BJsales.lead[1:130]
test_bjsales_lead <- BJsales.lead[131:150]

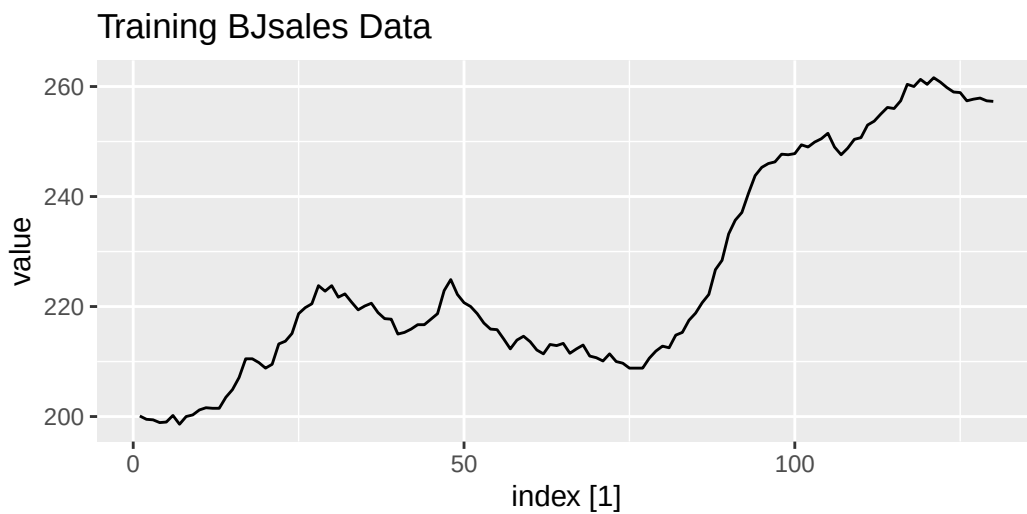
# Converting the datasets to tsibble
train_bjsales_ts <- as_tsibble(as.ts(train_bjsales))
test_bjsales_ts <- as_tsibble(as.ts(test_bjsales))
```

```

train_bjsales_lead_ts <- as_tsibble(as.ts(train_bjsales_lead))
test_bjsales_lead_ts <- as_tsibble(as.ts(test_bjsales_lead))

# Plotting the datasets
autoplot(train_bjsales_ts, value) +
  labs(title = "Training BJsales Data")

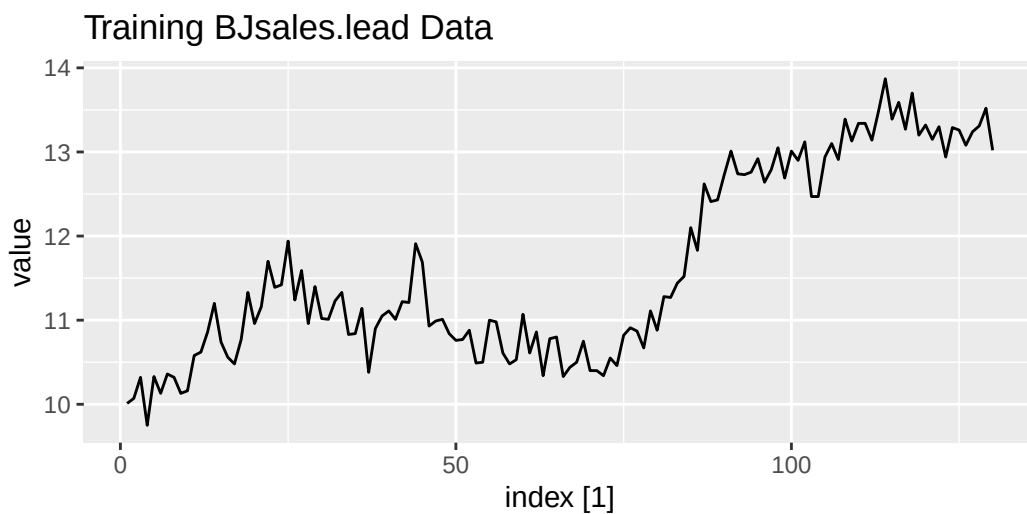
```



```

autoplot(train_bjsales_lead_ts, value) +
  labs(title = "Training BJsales.lead Data")

```



Part 2

For the BJSales series, remove the trend with any method you choose. Plot the resulting series and comment on what you see.

Comment on the plot:

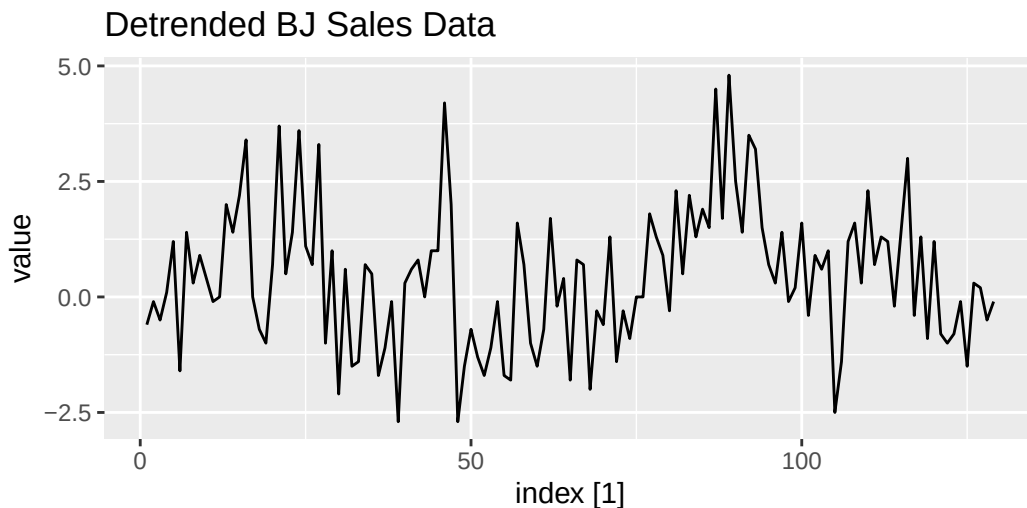
After removing the trend in the series by differencing, it still shows a pattern in the series, which indicate that there still exists trend in the series. However, the data appears to be more stationary in comparison with the original data. Additionally, the data is not centered at zero and has some spikes and sudden drops in the data indicating changing variance over time.

Hence, took second order differencing to make the series stationary, and now its better and does not exhibits any trend in the series.

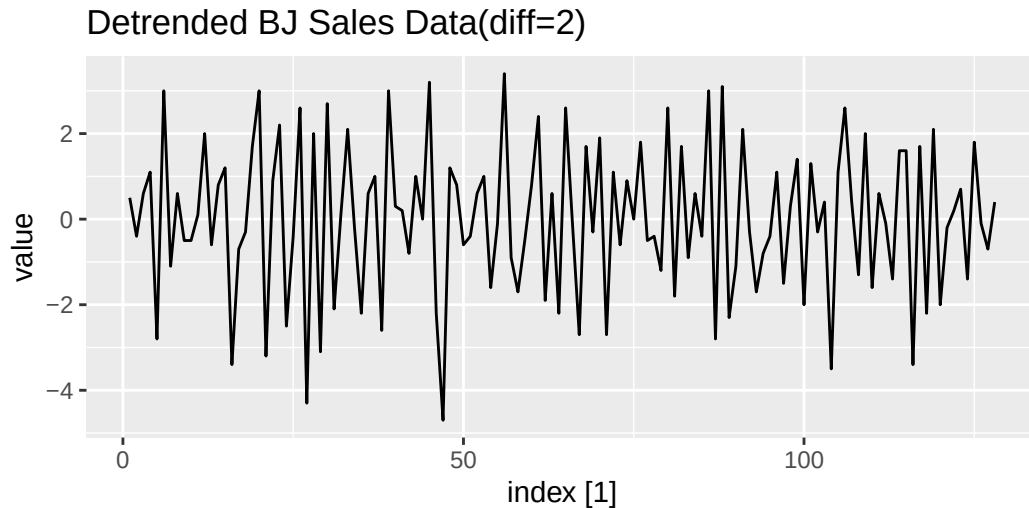
```
# Removing trend using differencing
bjsales_diff <- diff(train_bjsales)
bjsales_diff_ts <- as_tsibble(as.ts(bjsales_diff))

# With 2 differencing
bjsales_diff_2 <- diff(train_bjsales,
                      differences = 2)
bjsales_diff_2_ts <- as_tsibble(as.ts(bjsales_diff_2))

# Plotting the detrend dataset of BJSales
autoplot(bjsales_diff_ts, value) +
  labs(title = "Detrended BJ Sales Data")
```



```
# Plotting the detrend dataset of BJsales(diff=2)
autoplot(bjsales_diff_2_ts, value) +
  labs(title = "Detrended BJ Sales Data(diff=2)")
```



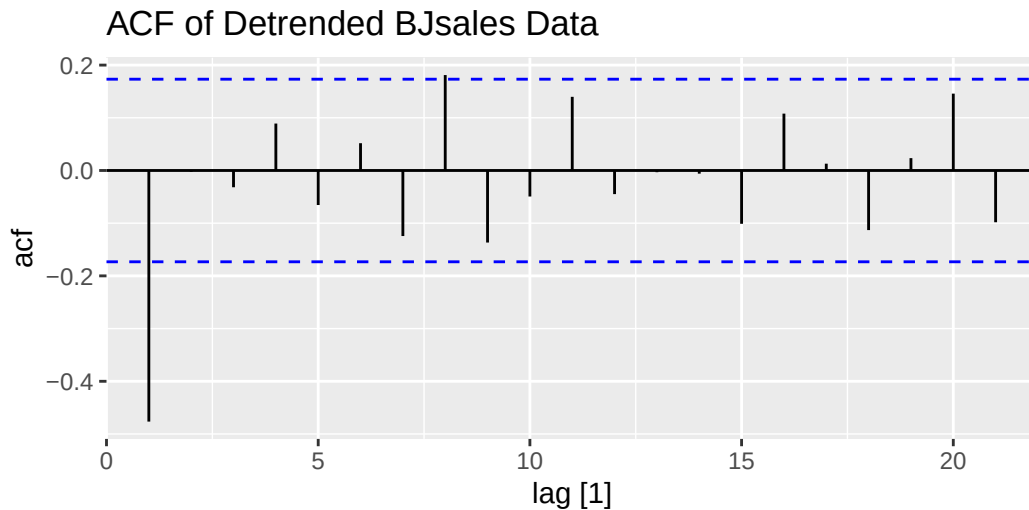
Part 3

Construct an ACF plot of the detrended BJSales series. Comment on what you see.

Comment on the plot:

The ACF plot shows, the first lag falls beyond the confidence interval indicating there is a positive autocorrelation in the data. The lag 7 is barely touching the confidence line. But, looking at the overall lags, it falls within the confidence line, indicating there is no autocorrelation and mostly resembles white noise. The detrend with second order differencing was effective.

```
# ACF plot of the detrended BJ Sales data
ACF(bjsales_diff_2_ts, bjsales_diff_2) |>
  autoplot() +
  ggtitle("ACF of Detrended BJsales Data")
```



Part 4

Fit a regression model for BJsales (without ARIMA errors) using BJsales.lead as a predictor variable. Examine the residuals of this fit and comment on the appropriateness of this regression model.

Comment on the plot:

The residual plot has a clear indication of autocorrelation. The histogram of residuals are slightly left skewed and are not symmetric. The pattern of the residuals looks random. Most of the lags in the acf plot are above the confidence showing significant autocorrelation present in the data. The residuals are not white noise. This suggests that regression model with ARIMA would be more appropriate.

```
# Fitting a regression model for BJsales without ARIMA
train_data <- tsibble(
  time = 1:130,
  sales = as.numeric(BJsales[1:130]),
  lead = as.numeric(BJsales.lead[1:130]),
  index = time
)

test_data <- tsibble(
  time = 131:150,
  sales = as.numeric(BJsales[131:150]),
  lead = as.numeric(BJsales.lead[131:150]),
```

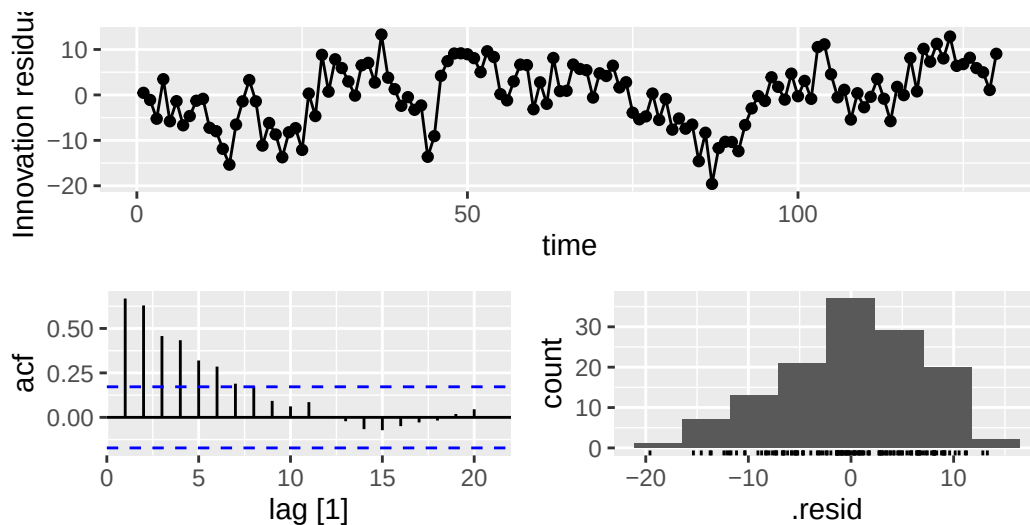
```

    index = time
  )

fit_lm <- model(
  train_data,
  TSLM(sales ~ lead)
)

fit_lm |>
  gg_tsresiduals()

```



Part 5

Fit another regression model for BJSales but with ARIMA errors using BJSales.lead as a predictor variable. Examine the residuals of this fit and comment on the appropriateness of this regression model.

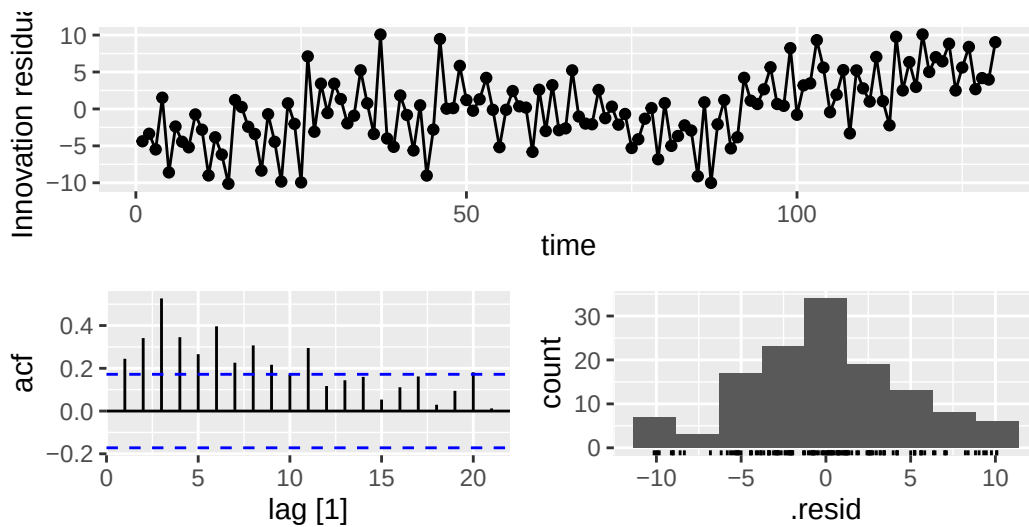
Comment on the plot:

The residuals look better but the upward trend in the later time is still shown. The ACF plot still shows the presence of autocorrelation as many lags are still beyond the confidence line. The histogram of the residuals are better and is quite symmetric, but can see a little right skewed plot. Overall, it indicates that the model has not fully captured the data structure.

```
#Fitting regression model with ARIMA
fit_arima <- model(
  train_data,
  ARIMA(sales ~ lead)
)

fit_arima
# # A mable: 1 x 1
#       `ARIMA(sales ~ lead)`
#               <model>
# 1 <LM w/ ARIMA(0,0,2) errors>

fit_arima |>
  gg_tsresiduals()
```



Part 6

Forecast the models from both parts 4 and 5 for the next 20 time points. Plot both forecasts on the same plot.

```
fc_bjsales_lm <- forecast(fit_lm, new_data = test_data)

p1 <- autoplot(fc_bjsales_lm, train_data) +
  autolayer(test_data, sales) +
```

```

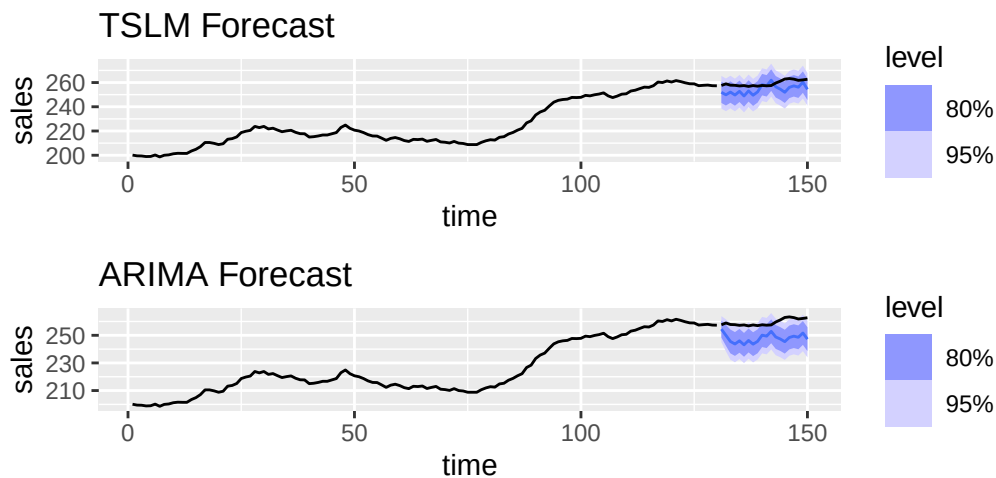
labs(title = "TSLM Forecast")

fc_bjsales_arima <- forecast(fit_arima, new_data = test_data)

p2 <- autoplot(fc_bjsales_arima, train_data) +
  autolayer(test_data, sales) +
  labs(title = "ARIMA Forecast")

p1/p2

```



Part 7

Comment on the differences between the forecasts of the two models you see on the plot from part 6.

Comment on the plot:

Almost both the model did a good job in predicting. The forecast of the TSLM model does not consider the past sales pattern. The confidence interval is wider, indicating more uncertainty. TSLM are generally less responsive to the recent changes. Whereas the ARIMA model forecast follows the recent trend better. The confidence intervals range is a bit narrow when compared with the TSLM indicating more confident and less uncertainty. It tends to be more accurate than TSLM.